EXPERIMENT 6A: - EXPLORATORY DATA ANALYSIS

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Learning Objective

To perform exploratory data analysis on a real-world Netflix dataset using Python.

Problem Statement

Netflix wants to analyze its movies and TV shows to understand content distribution, ratings, genres, countries, and trends over time.

TASKS TO PERFORM

1. Download the Netflix titles dataset from Kaggle and convert it into a DataFrame.
2. Explore the dataset by checking its structure, data types, and missing values.
3. Content Distribution Analysis:
    - How many movies and TV shows are in the dataset?
    - What is the distribution of ratings (e.g., TV-MA, TV-PG, R)?
    - Which countries have the most Netflix content?
4. Trend Analysis:
    - In which years were the most shows and movies released?
    - In which years were titles added to Netflix?
    - How has Netflix's content trend changed over time?
5. Duration and Content Type Analysis:
    - What is the distribution of movie durations in minutes?
    - How many TV shows have multiple seasons?
6. Genre Analysis:
    - What are the top 10 most common genres on Netflix?
    - Which directors have the most titles on Netflix?
    - Which actors appear most frequently across Netflix shows and movies?
7. Text and Description Analysis:
    - What are the most common words in descriptions? (Word cloud)
8. Which months of the year have the most content added?
9. Which countries produce more movies versus TV shows?

10. What are the most common ratings for movies and TV shows?
11. What are the oldest and newest titles available on Netflix?
12. Which genres frequently co-occur together? (Genre correlation heatmap)

AIM

To perform exploratory data analysis on the Netflix dataset and identify meaningful patterns and trends in its content.

ALGORITHM

1. Load the Netflix dataset and inspect its structure.
2. Check missing values and clean the dataset.
3. Analyze content types, ratings, genres, countries, and release years.
4. Visualize trends using suitable charts.
5. Identify common genres, directors, actors, and content trends.
6. Analyze descriptions using word frequency and genre correlations.

PROGRAM AND OUTPUT

Question 1: Download the Netflix titles dataset from the internet and convert it to a DataFrame.

```
# LUQMAN T (241801141)

import pandas as pd

url = "https://raw.githubusercontent.com/prasertcbs/basic-dataset/master/netflix_titles.csv"
df = pd.read_csv(url)

print(df.head())
print("Shape:", df.shape)
```

```
    show_id     type                                  title  \
0  81145628    Movie  Norm of the North: King Sized Adventure
1  80117401    Movie               Jandino: Whatever it Takes
2  70234439  TV Show                   Transformers Prime
3  80058654  TV Show         Transformers: Robots in Disguise
4  80125979    Movie                           #realityhigh

                director  \
0  Richard Finn, Tim Maltby
1                     NaN
2                     NaN
3                     NaN
4         Fernando Lebrija

                                        cast  \
0  Alan Marriott, Andrew Toth, Brian Dobson, Cole...
1                                Jandino Asporaat
2  Peter Cullen, Sumalee Montano, Frank Welker, J...
3  Will Friedle, Darren Criss, Constance Zimmer, ...
4  Nesta Cooper, Kate Walsh, John Michael Higgins...

                              country         date_added  release_year  \
0  United States, India, South Korea, China  September 9, 2019          2019
1                           United Kingdom  September 9, 2016          2016
2                            United States  September 8, 2018          2013
3                            United States  September 8, 2018          2016
4                            United States  September 8, 2017          2017

     rating  duration                          listed_in  \
0     TV-PG    90 min  Children & Family Movies, Comedies
1     TV-MA    94 min                   Stand-Up Comedy
2  TV-Y7-FV  1 Season                          Kids' TV
3     TV-Y7  1 Season                          Kids' TV
4     TV-14    99 min                          Comedies

                                      description
0  Before planning an awesome wedding for his gra...
1  Jandino Asporaat riffs on the challenges of ra...
2  With the help of three human allies, the Autob...
3  When a prison ship crash unleashes hundreds of...
4  When nerdy high schooler Dani finally attracts...
Shape: (6234, 12)
```

Question 2: Explore the Netflix dataset.

```
# LUQMAN T (241801141)

print(df.info())
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
print("\nSummary:\n", df.describe(include="all"))
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6234 entries, 0 to 6233
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       6234 non-null   int64
 1   type          6234 non-null   object
 2   title         6234 non-null   object
 3   director      4265 non-null   object
 4   cast          5664 non-null   object
 5   country       5758 non-null   object
 6   date_added    6223 non-null   object
 7   release_year  6234 non-null   int64
 8   rating        6224 non-null   object
```

```
 9   duration      6234 non-null   object
 10  listed_in     6234 non-null   object
 11  description   6234 non-null   object
dtypes: int64(2), object(10)
memory usage: 584.6+ KB
None

Missing Values:
 show_id            0
type                0
title               0
director         1969
cast              570
country           476
date_added         11
release_year        0
rating             10
duration            0
listed_in           0
description         0
dtype: int64

Duplicate Rows: 0

Summary:
              show_id   type        title               director  \
count   6.234000e+03   6234         6234                   4265
unique           NaN      2         6172                   3301
top              NaN  Movie  The Silence  Raúl Campos, Jan Suter
freq             NaN   4265            3                     18
mean    7.670368e+07    NaN          NaN                    NaN
std     1.094296e+07    NaN          NaN                    NaN
min     2.477470e+05    NaN          NaN                    NaN
25%     8.003580e+07    NaN          NaN                    NaN
50%     8.016337e+07    NaN          NaN                    NaN
75%     8.024489e+07    NaN          NaN                    NaN
max     8.123573e+07    NaN          NaN                    NaN

                       cast        country       date_added  release_year  \
count                  5664           5758             6223    6234.00000
unique                 5469            554             1524           NaN
top     David Attenborough  United States  January 1, 2020           NaN
freq                     18           2032              122           NaN
mean                    NaN            NaN              NaN   2013.35932
```

Question 3: Perform Content Distribution Analysis: (a) Movies vs TV Shows, (b) ratings distribution, and (c) countries with the most content.

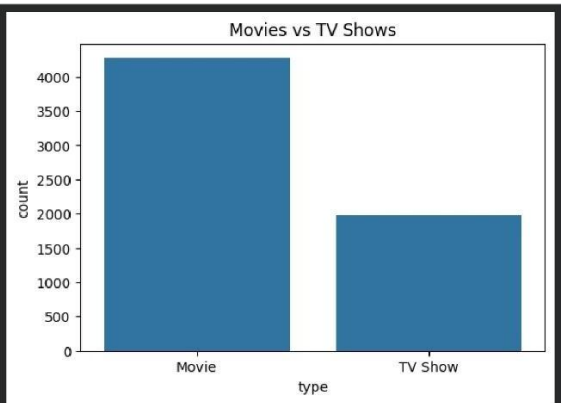
```
# LUQMAN T (241801141)

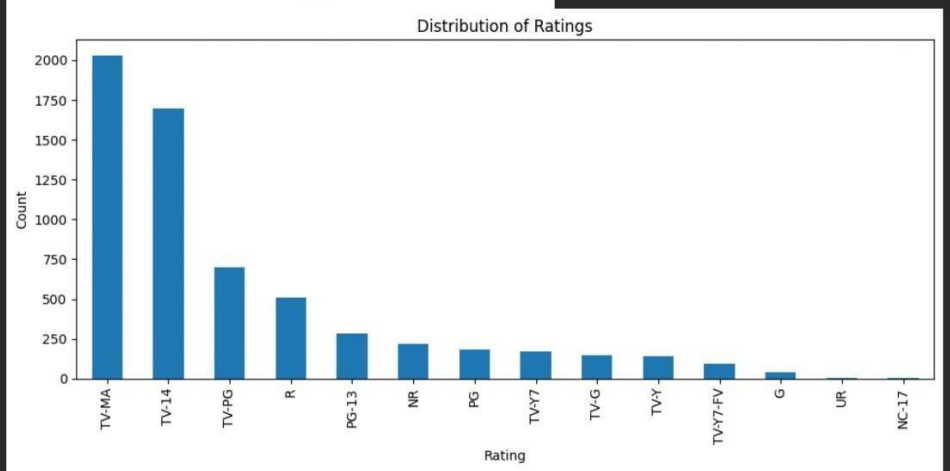
import matplotlib.pyplot as plt
import seaborn as sns

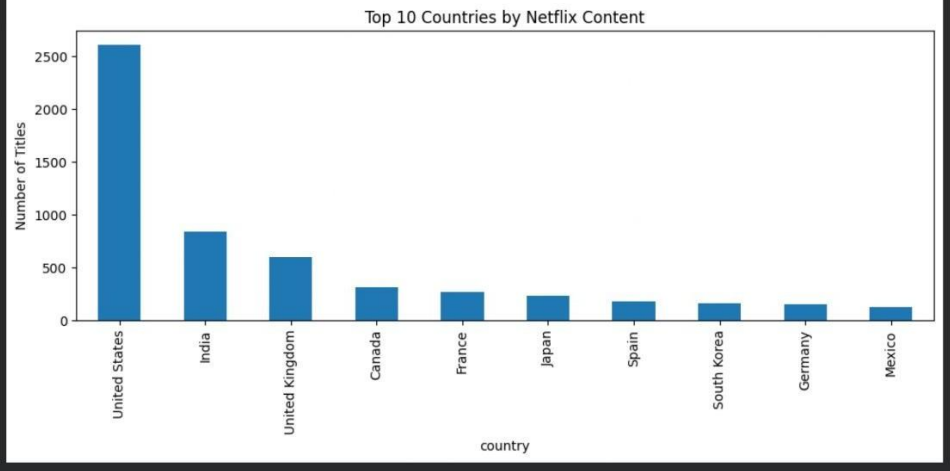
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="type")
plt.title("Movies vs TV Shows")
plt.show()

plt.figure(figsize=(10, 5))
df["rating"].value_counts().plot(kind="bar")
plt.title("Distribution of Ratings")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

countries = df["country"].dropna().str.split(", ").explode()
plt.figure(figsize=(10, 5))
countries.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Countries by Netflix Content")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()
```







Question 4: Perform Time-Based Trend Analysis: (a) release years with most titles, (b) years titles were added, and (c) content growth over time.

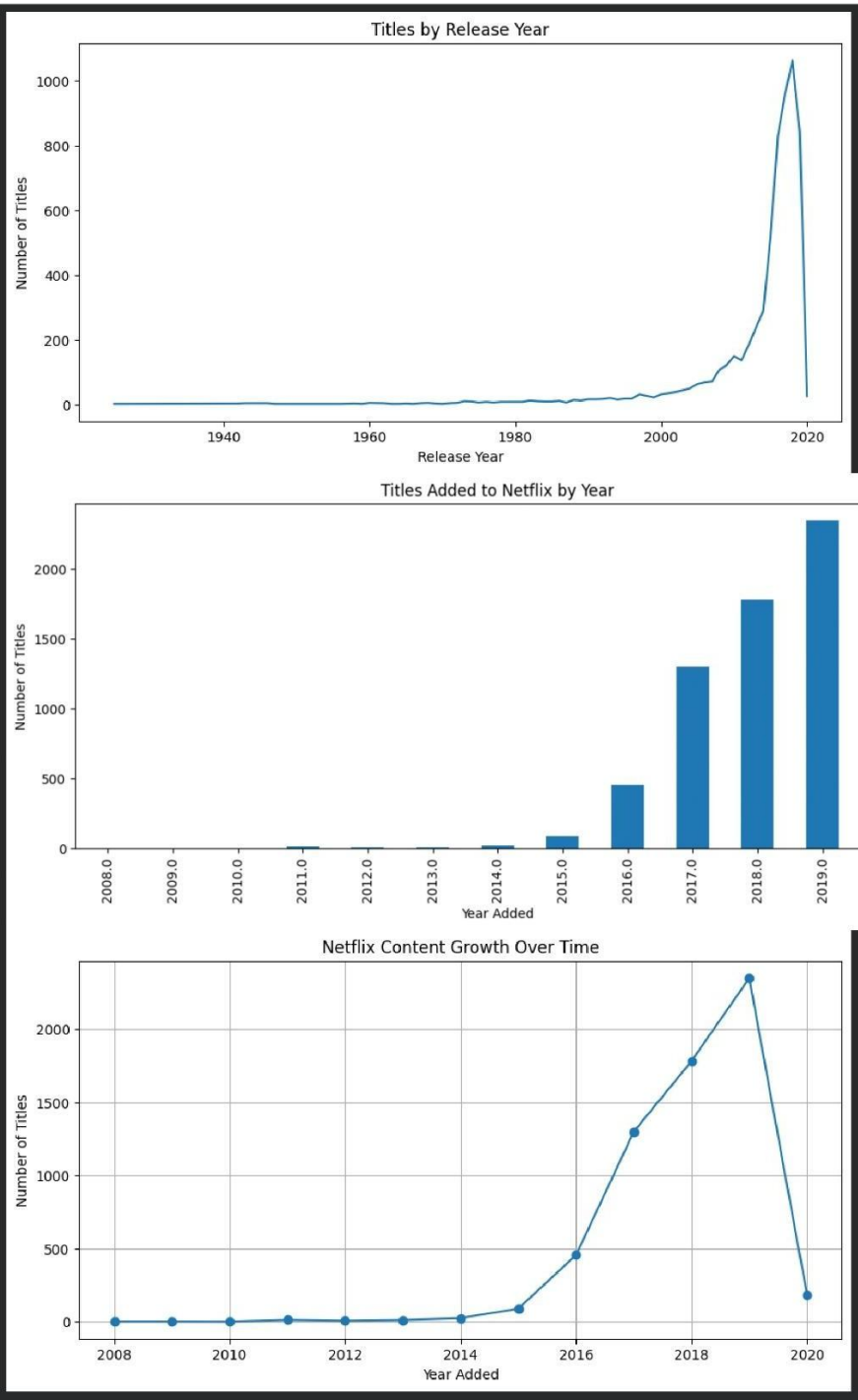
```
# LUQMAN T (241801141)

df["date_added"] = pd.to_datetime(df["date_added"].astype(str).str.strip(), errors="coerce")
df["release_year"] = pd.to_numeric(df["release_year"], errors="coerce")

plt.figure(figsize=(10, 5))
df["release_year"].value_counts().sort_index().plot()
plt.title("Titles by Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()

added_year = df["date_added"].dt.year.value_counts().sort_index()
plt.figure(figsize=(10, 5))
added_year.plot(kind="bar")
plt.title("Titles Added to Netflix by Year")
plt.xlabel("Year Added")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()

plt.figure(figsize=(10, 5))
added_year.plot(kind="line", marker="o")
plt.title("Netflix Content Growth Over Time")
plt.xlabel("Year Added")
plt.ylabel("Number of Titles")
plt.grid(True)
plt.show()
```



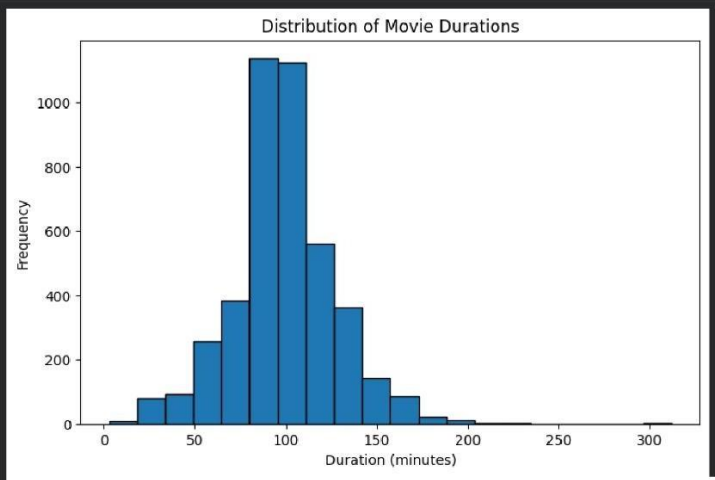
Question 5: Perform Duration and Content Type Analysis: (a) movie duration distribution and (b) number of TV shows by seasons.

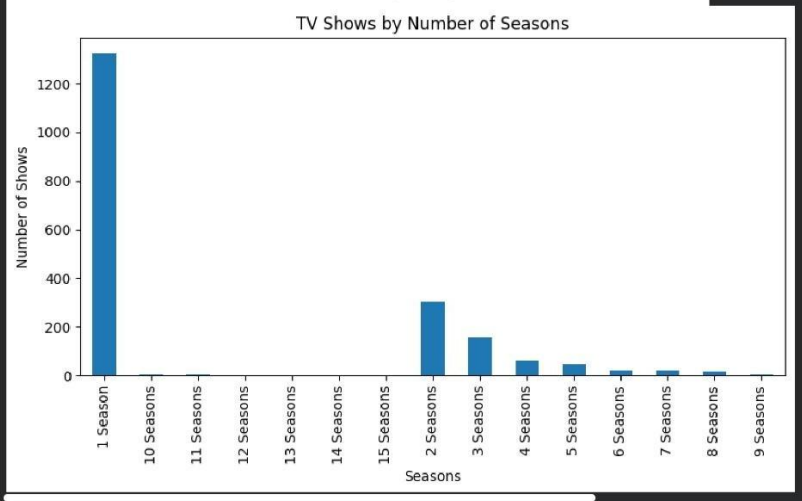
```
# LUQMAN T (241801141)

df["duration_value"] = pd.to_numeric(
    df["duration"].str.extract(r"(\d+)")[0], errors="coerce"
)
movies = df[df["type"] == "Movie"]
shows = df[df["type"] == "TV Show"]

plt.figure(figsize=(8, 5))
movies["duration_value"].dropna().plot(kind="hist", bins=20, edgecolor="black")
plt.title("Distribution of Movie Durations")
plt.xlabel("Duration (minutes)")
plt.show()

plt.figure(figsize=(8, 5))
shows["duration"].value_counts().sort_index().plot(kind="bar")
plt.title("TV Shows by Number of Seasons")
plt.xlabel("Seasons")
plt.ylabel("Number of Shows")
plt.tight_layout()
plt.show()
```





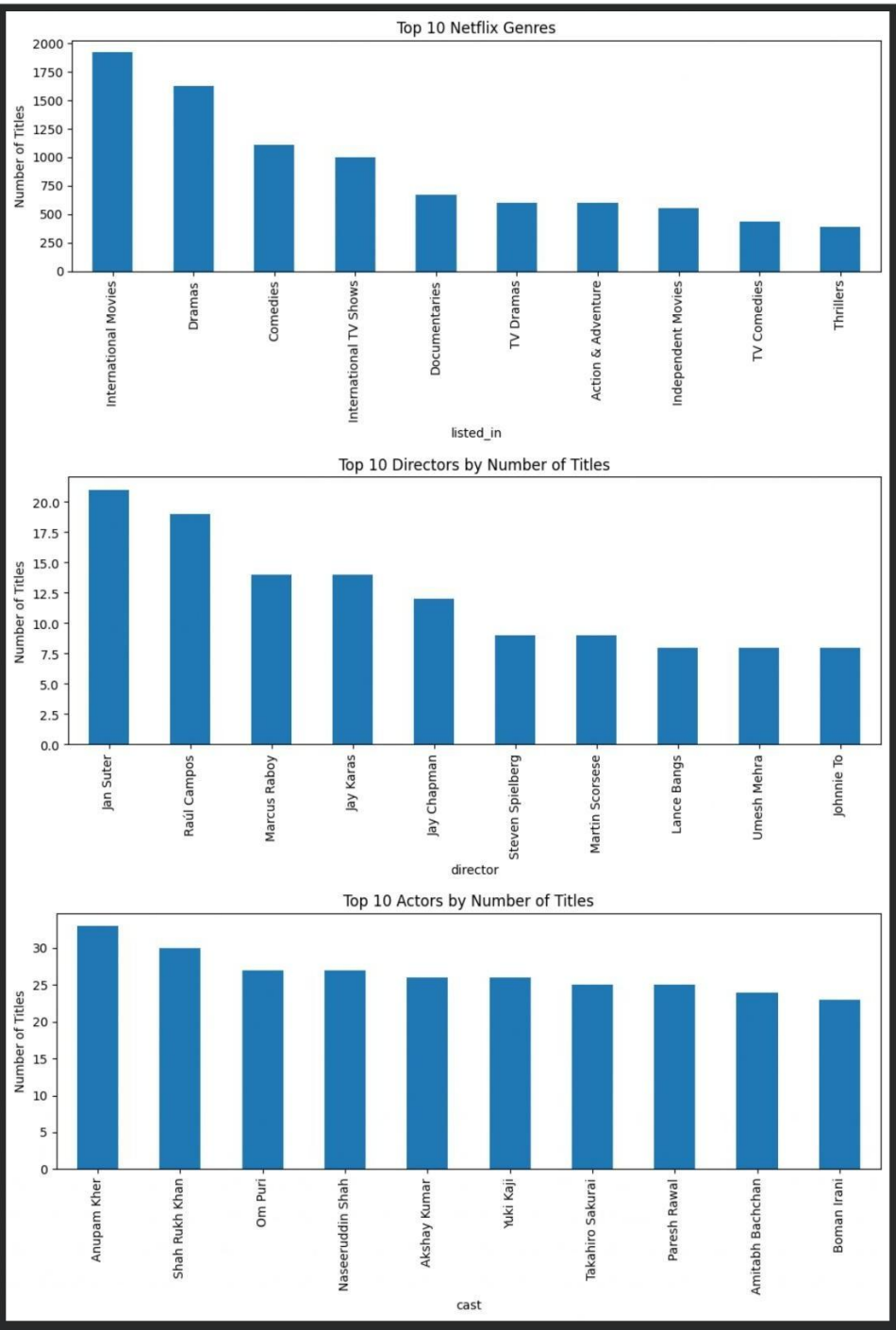
Question 6: Perform Genre and Categories Analysis: (a) top 10 genres, (b) directors with most titles, and (c) most frequent actors.

```
# LUQMAN T (241801141)

genres = df["listed_in"].dropna().str.split(", ").explode()
plt.figure(figsize=(10, 5))
genres.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Netflix Genres")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()

directors = df["director"].dropna().str.split(", ").explode()
plt.figure(figsize=(10, 5))
directors.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Directors by Number of Titles")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()

actors = df["cast"].dropna().str.split(", ").explode()
plt.figure(figsize=(10, 5))
actors.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Actors by Number of Titles")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()
```


Top 10 Netflix Genres
Number of Titles
2000
1750
1500
1250
1000
750
500
250
0
International Movies
Dramas
Comedies
International TV Shows
Documentaries
TV Dramas
Action & Adventure
Independent Movies
TV Comedies
Thrillers
listed_in
Top 10 Directors by Number of Titles
Number of Titles
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Jan Suter
Raúl Campos
Marcus Raboy
Jay Karas
Jay Chapman
Steven Spielberg
Martin Scorsese
Lance Bangs
Umesh Mehra
Johnnie To
director
Top 10 Actors by Number of Titles
Number of Titles
30
25
20
15
10
5
0
Anupam Kher
Shah Rukh Khan
Om Puri
Naseeruddin Shah
Akshay Kumar
Yuki Kaji
Takahiro Sakurai
Paresh Rawal
Amitabh Bachchan
Boman Irani
cast

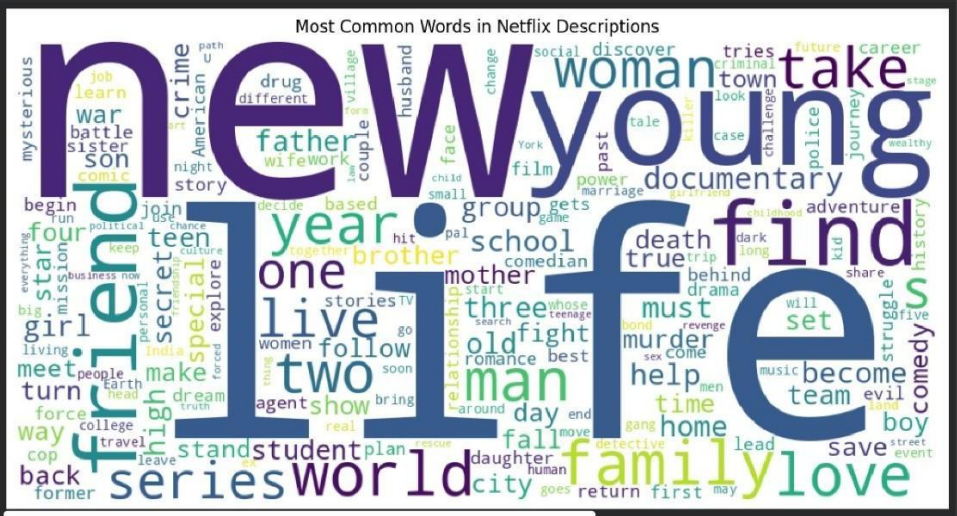
Question 7: Find the most common words in the description field and display them as a word cloud.

```
# LUQMAN T (241801141)

from wordcloud import WordCloud, STOPWORDS

text = " ".join(df["description"].dropna().astype(str))
wordcloud = WordCloud(
    width=1000, height=500, background_color="white",
    stopwords=STOPWORDS, collocations=False
).generate(text)

plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.title("Most Common Words in Netflix Descriptions")
plt.show()
```

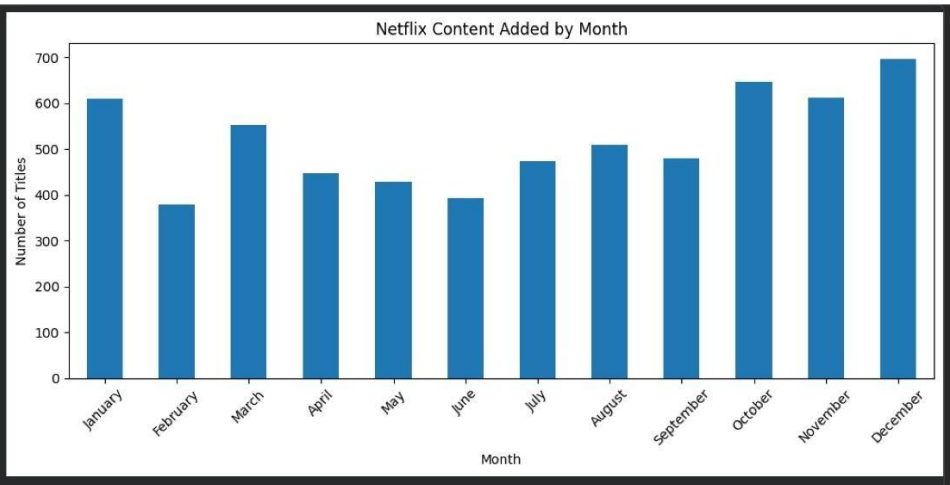


Question 8: Find the months of the year in which Netflix adds the most content.

```
# LUQMAN T (241801141)

month_names = df["date_added"].dropna().dt.month_name()
month_order = [
    "January", "February", "March", "April", "May", "June",
    "July", "August", "September", "October", "November", "December"
]

plt.figure(figsize=(10, 5))
month_names.value_counts().reindex(month_order).plot(kind="bar")
plt.title("Netflix Content Added by Month")
plt.xlabel("Month")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



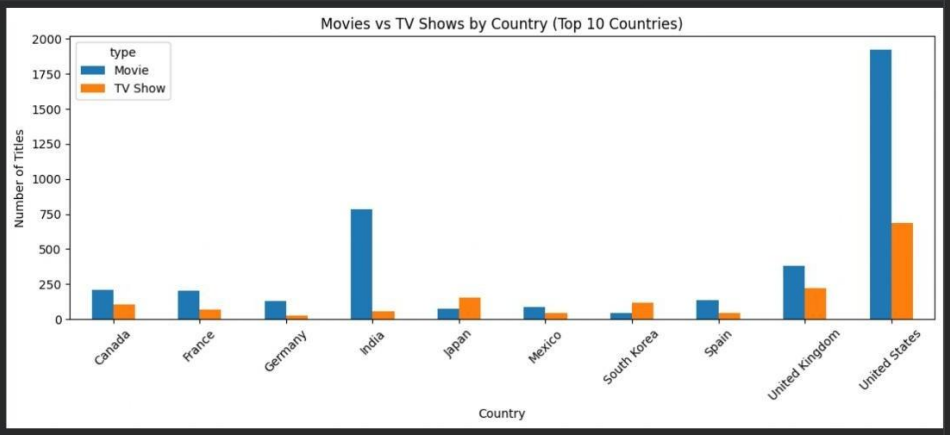
Question 9: Compare the number of movies and TV shows produced by each country.

```
# LUQMAN T (241801141)

country_type = df[["type", "country"]].dropna().copy()
country_type["country"] = country_type["country"].str.split(", ")
country_type = country_type.explode("country")

top_countries = country_type["country"].value_counts().head(10).index
comparison = country_type[country_type["country"].isin(top_countries)]

table = comparison.groupby(["country", "type"]).size().unstack(fill_value=0)
table.plot(kind="bar", figsize=(11, 5))
plt.title("Movies vs TV Shows by Country (Top 10 Countries)")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



RESULT

Exploratory data analysis was performed on the Netflix dataset to identify content distributions, release trends, popular genres, and relationships among content categories.